

SAINSHAND, Mongolia

WMO: 443540

Lat: 44.868N

Lon: 110.117E

Elev: 3077

StdP: 13.13

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.9	-13.3	-25.4	1.5	-11.6	-21.6	1.9	-10.1	34.8	4.4	30.5	3.1	7.9	320	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	94.5	66.0	91.2	64.8	87.9	63.3	70.0	87.8	67.7	84.2	65.9	81.8	10.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.7	102.6	77.3	62.4	94.5	74.7	60.2	87.3	73.7	36.1	87.6	34.1	83.9	32.5	82.0	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.6	27.1	23.2	-20.8	100.0	5.2	2.9	-24.5	102.1	-27.6	103.8	-30.5	105.4	-34.3	107.5		
(5)			-20.7	72.6	5.6	3.1	-24.7	74.8	-28.0	76.6	-31.1	78.3	-35.2	80.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.5	1.5	11.5	28.7	46.1	59.1	70.6	76.3	72.1	60.3	42.8	21.4	5.8		
	DBStd	27.29	8.43	9.89	10.44	9.56	9.05	6.97	5.77	6.12	8.21	8.98	10.86	8.40		
	HDD50	5948	1503	1077	661	185	23	0	0	0	15	255	859	1370		
	HDD65	9437	1968	1497	1124	569	227	25	1	11	183	688	1309	1835		
	CDD50	2849	0	0	2	67	305	619	814	686	324	32	0	0		
	CDD65	864	0	0	0	1	45	194	350	232	42	0	0	0		
	CDH74	9200	0	0	0	40	626	2108	3794	2223	403	6	0	0		
	CDH80	3707	0	0	0	6	204	850	1693	854	99	1	0	0		
	WSAvg	10.0	8.9	9.5	11.2	12.2	12.9	10.6	9.2	8.4	9.2	9.6	9.2	9.3		
	PrecAvg	4.3	0.0	0.0	0.1	0.1	0.3	0.7	1.2	1.1	0.4	0.2	0.1	0.1		
	PrecMax	9.0	0.2	0.4	0.4	0.6	1.3	2.9	5.4	4.1	1.6	1.0	0.6	1.6		
(15) Precipitation	PrecMin	2.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0		
	PrecStd	1.7	0.0	0.1	0.1	0.1	0.3	0.6	0.9	0.9	0.4	0.2	0.1	0.2		
	0.4%	DB	30.0	42.1	63.8	79.0	91.8	97.2	100.5	96.6	87.5	72.4	52.8	31.9		
	0.4%	MCWB	23.9	32.2	43.6	53.9	60.1	65.5	68.8	65.8	61.9	53.1	39.8	26.4		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	22.3	36.6	56.3	73.2	85.5	92.3	95.6	91.8	82.7	66.5	46.2	26.0		
	2%	MCWB	17.9	28.4	39.6	50.3	57.1	63.2	67.6	65.5	59.2	49.1	35.9	21.1		
	5%	DB	18.1	31.6	50.7	68.0	81.0	88.3	92.1	88.1	78.7	61.7	41.5	21.6		
	5%	MCWB	14.5	25.3	37.0	47.7	55.5	62.5	66.2	64.0	57.5	46.3	32.7	17.8		
	10%	DB	14.2	27.1	45.6	63.0	76.5	84.9	88.6	84.6	74.6	57.3	36.8	18.0		
	10%	MCWB	11.4	21.9	33.7	45.4	53.3	60.8	64.6	62.3	55.6	43.7	29.7	14.9		
	0.4%	WB	24.2	33.6	44.4	56.2	62.7	69.3	74.4	71.9	65.7	54.9	40.6	26.1		
	0.4%	MCDB	30.3	40.7	62.1	75.0	86.2	90.6	95.2	88.1	80.7	69.6	51.3	31.6		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	18.3	28.9	40.8	51.3	59.0	66.7	71.2	69.0	62.0	50.2	36.2	21.5		
	2%	MCDB	22.4	36.6	55.7	70.6	80.4	86.6	89.5	84.2	76.3	64.9	46.0	25.9		
	5%	WB	14.8	25.5	37.3	48.5	56.6	64.6	68.7	66.7	59.4	46.9	33.1	18.1		
	5%	MCDB	18.2	31.8	50.5	66.7	77.2	82.4	86.1	81.2	74.0	60.0	41.5	21.6		
	10%	WB	11.7	22.4	34.1	45.7	54.2	62.7	66.8	64.7	57.1	44.2	30.2	15.2		
	10%	MCDB	14.4	27.6	45.6	62.6	73.9	79.9	82.7	79.6	71.8	56.8	37.1	18.1		
(35) Mean Daily Temperature Range	MDBR	17.1	21.2	23.1	23.6	23.3	21.0	19.6	19.9	21.4	20.9	18.4	16.0			
	5% DB	MCDBR	21.0	25.7	28.5	29.8	29.0	25.3	23.7	23.8	25.9	25.4	23.0	19.3		
	5% DB	MCWBR	17.4	19.4	18.0	17.4	15.2	11.8	10.2	10.5	12.7	15.1	15.9	15.4		
	5% WB	MCDBR	20.7	25.2	27.9	27.8	26.2	22.1	20.6	20.8	21.9	24.1	22.7	19.1		
	5% WB	MCWBR	17.3	19.1	18.0	17.0	15.3	11.4	10.8	11.1	13.0	15.3	16.0	15.5		
(40) Clear-Sky Solar Irradiance	taub	0.261	0.285	0.323	0.377	0.421	0.450	0.419	0.397	0.362	0.316	0.276	0.259			
	taud	2.324	2.263	2.197	2.086	2.008	2.017	2.173	2.209	2.237	2.303	2.327	2.306			
	Ebn at Noon	274	285	287	279	269	261	268	270	271	269	264	261			
	Edh at Noon	24	32	39	48	54	54	45	42	38	30	24	23			
(44) All-Sky Solar Radiation	RadAvg	658	1017	1546	1958	2194	2184	2106	1925	1596	1166	775	557			
	RadStd	72	74	69	88	101	117	88	63	62	47	44	59			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.59	N/A	N/A	N/A	-1.53	-1.48	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	-2.09	N/A	-1.12	-1.40	N/A	N/A	N/A	N/A

Nomenclature: See separate page